

MOHAK SINGH RANA

Junior Undergraduate

Department of Computer Science and Engineering

Indian Institute of Technology Kanpur

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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	9.4/10
2021	CBSE (XII)	Mangalam D.A.V. Public School, Kota	97.6%
2019	CBSE (X)	Mangalam D.A.V. Public School, Kota	97.6%

SCHOLASTIC ACHIEVEMENTS

- Recipient of the **Academic Excellence Award** twice for exceptional academic performance at **IIT Kanpur** (2021, 2022)
- Secured **All India Rank 165** in **Joint Entrance Examination Advanced** among 1,50,000+ candidates (2021)
- Secured **All India Rank 70** in **JEE Mains** with **100 percentile in Mathematics** among 10,00,000+ candidates (2021)
- Merit awardee** and **Rajasthan State Topper** in Indian Olympiad Qualifiers Physics (**IOQP**) conducted by **IAPT** (2021)
- Secured **All India Rank 165** in **KVPY (SX Stream)** and was awarded **KVPY Fellowship** by IISc Bangalore (2021)
- Secured **All India Rank 45** in **KVPY (SA Stream)** and was awarded **KVPY Fellowship** by IISc Bangalore (2020)
- Qualified Regional Mathematical Olympiad (**RMO**), leading to **INMO** (among top 30 students in **Rajasthan** state) (2019)
- Qualified National Standard Examination in Junior Science (**NSEJS**), leading to **INJSO** (top 300 among 80,000+) (2019)
- Qualified for Indian National Astronomy Olympiad (**INAO**) conducted by **HBCSE** (top 300 among 80,000+ students) (2019)
- Recipient of the prestigious National Talent Search Examination (**NTSE**) **scholarship** conducted by NCERT (2019)
- Awarded **Certificate of Merit** for scoring **100/100** in Mathematics, Science and Social Science in **AISSE** (**top 0.1%**) (2019)

WORK EXPERIENCE

Low Latency, Low Loss and Scalable Throughput (L4S)

(May'23-Jul'23)

Research Intern at **New York University** | Prof. Shivendra Panwar, Prof. Fraida Fund

Objective	- Performance of L4S when deployed on the global network and its coexistence with the existing traffic - Impacts on parameters like latency, bandwidth, etc by factors like congestion control algorithm, ECN, etc
Approach	- Conducted TCP Congestion Control experiment depicting classic sawtooth pattern of congestion window - Analysed the performance of TCP Reno, Cubic & Vegas individually & when sharing the same bottleneck - Experimented progressively from classic TCP flow to AQM flow (varying ECN threshold) to the L4S flow - Examined latency, bandwidth and other parameters in a network coexisting of L4S and classic traffic
Impact	- Conducted experiment showing the scalable sawtooth behaviour of L4S and working of enhanced ECN - Demonstrated the coexistence of the L4S traffic & the classic traffic using Dual-Queue Coupled AQM

PROJECTS

Intrusion in IoT Devices

(Dec'22-Ongoing)

- Aimed to exploit the **vulnerabilities** of Bluetooth communication & then develop a **intrusion detection system (IDS)**
- Performed denial of services attack (**DoS attack**) using **Bluetooth-DOS-Script** & analysed pcap files using **Wireshark**
- Conducted man-in-the-middle (**MITM**) attack using **Btlejuice** to setup attacker's host & proxy, & on-the-fly data breach

Event-Split

(Jan'23-May'23)

- Designed a **web application** for managing expenses and calendars using **ReactJS**, **NodeJS**, **ExpressJS**, and **MongoDB**
- Implemented **simplify function** using **binary heap** returning the minimum number of transactions needed to settle a group
- Documented** software formally (SRS, SDD, SID, User Manual) and conducted **testing** to achieve an almost bug-free software

CSE Bubble

(Jan'23-May'23)

- Developed a **Verilog HDL processor** CSE.BUBBLE in accordance with the **MIPS-32** Instruction Set Architecture (**ISA**)
- Designed a finite state machine (**FSM**) for **control signals** to fetch instruction, decoding, arithmetic logic & memory write
- Validated the processor on **MIPS code** of **bubble sort** storing machine code in instruction memory & output in data memory

RELEVANT COURSES

Data Structure and Algorithms	Fundamentals of Computing	Computer Organisation
Discrete Mathematics	Logic for Computer Science	Probability for Computer Science
Linear Algebra and ODE	Real Analysis and Multi-Variate Calculus	Software Development and Operations

TECHNICAL SKILLS

- Programming** - C, C++, Python, Verilog, MIPS
- Utilities** - Linux, Bash, Git, L^AT_EX, MERN stack
- Networks** - Wireshark, BtleJuice, CloudLab, FABRIC, Gattacker
- Web Development** - HTML, CSS, Javascript

POSITIONS OF RESPONSIBILITY

Assistant Placement Coordinator

(Mar'23-Present)

Leadership	- Coordinated internship process of 1000+ students & placement process of 1500+ students in 3-tier team - Developing corporate liaisons with 50+ corporate giants for internship and placement drive 2023-24 - Leading a team of 70+ members responsible for coordinating career fest, internship and placement drive
Initiatives	Invited startups for recruitment to increase the success rate of institute internships and placements
Impact	Successfully attracted various startups to register for the career fest, internship drive & placement drive